

# The Effect of LIHTC on Housing Markets: Quantities, Rents, and Welfare

Fernando Ferreira, Gillian Meyer, Matthew Turner

Cody Cook

NBER Real Estate

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# The literature on “crowd-out” of LIHTC

| Paper                          | Crowd-out                         |
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| Ferreira, Meyer, Turner (2026) | 10%                               |
| Baum-Snow and Marion (2009)    | 20%                               |
| Sinai and Waldfogel (2005)     | 50-70%                            |
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| Soltas (2024)                  | 65-75%                            | 40 states,<br>2005-2019  | +10 years     | ITT/ATT  |
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Q: Suppose both true. Is there a plausible model that can rationalize both?

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**Note:** Soltas (2024) also tests with neighborhood-level and finds substantial crowd-out

## Setup & demand

**Setup.** 1 neighborhood, 200 parcels, 15 vacant

$N_{MR}$ ,  $N_{LIHTC}$  market-rate and LIHTC developers

**Demand.** Constant elasticity inverse demand:

$$r(H) = \bar{r} \left( \frac{H}{H_0} \right)^{-1/\eta}$$

All parcels charge same rent  $r$

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where cost  $C_j \sim \mathcal{N}(\mu_C, \sigma_C)$  and subsidy is  $S_j$

Decision follows nested logit

Nests: {don't build} and {vacant parcels}

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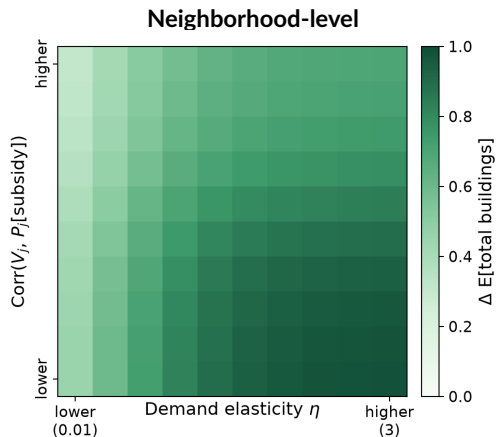
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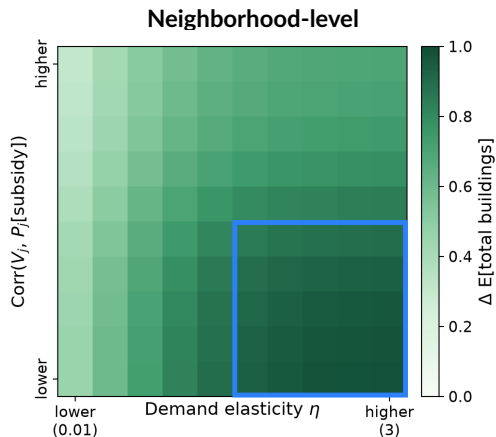
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# Crowd-out depends on demand elasticity + subsidy targeting

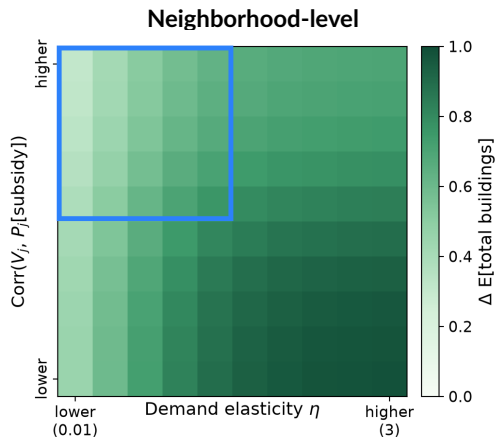


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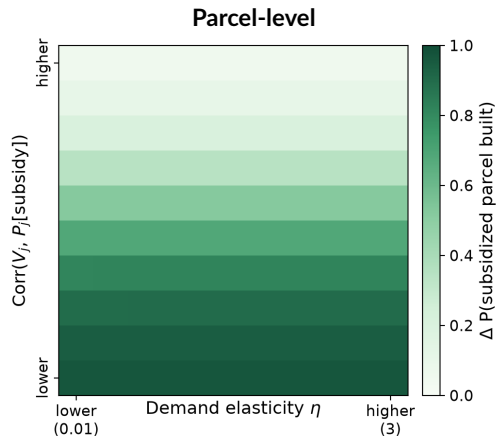
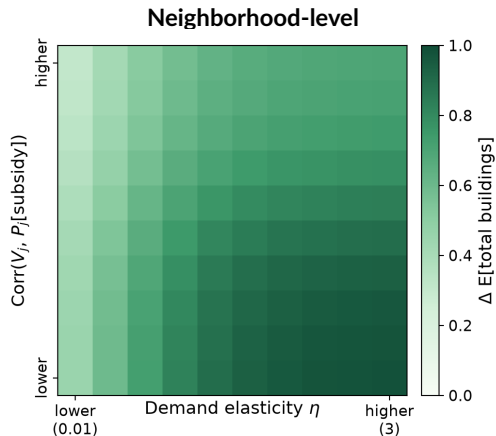
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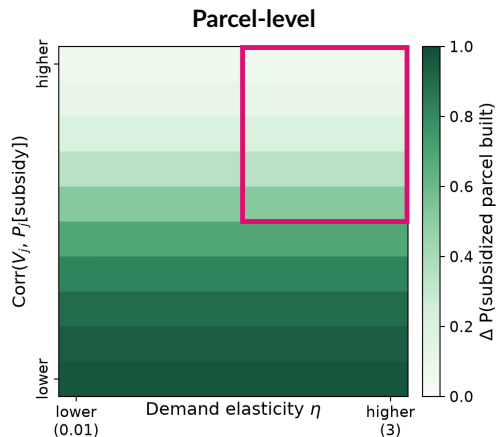
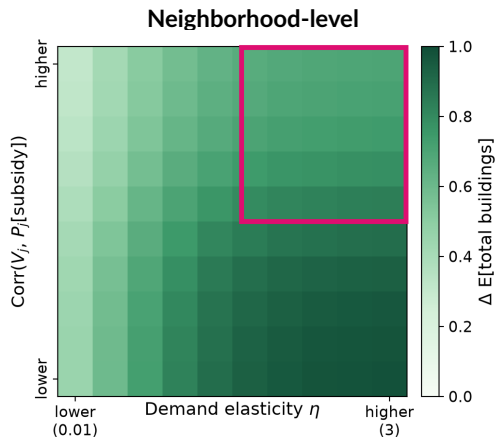


Low demand elast., high subsidy targeting  
 $\Rightarrow$  more crowd-out

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**Low neighborhood-level crowd-out & high parcel-level crowd-out!**

# Which world are we in?

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2. Correlation between parcel value & probability it gets subsidy
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- Testable. Do approved parcels look like ones that get developed for market-rate?

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- Testable. Vacant parcels nearby?

## Stepping back: what's the relevant definition of “market”?

Sinai and Waldfogel (2005)

*The ideal unit of observation would be a market area. A narrow geographic unit like a tract would be inappropriate because the private housing adjustment would likely be outside the tract* [slightly paraphrased]

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The usual test: look for effects on neighboring “markets”

- **This paper:** +6 units in neighboring block groups (std. error of 2.5 units). No effects on rents
- Other evidence that spatial spillovers are very local in Diamond and McQuade (2019); Asquith and Mast (2024); McCulloch (2026)

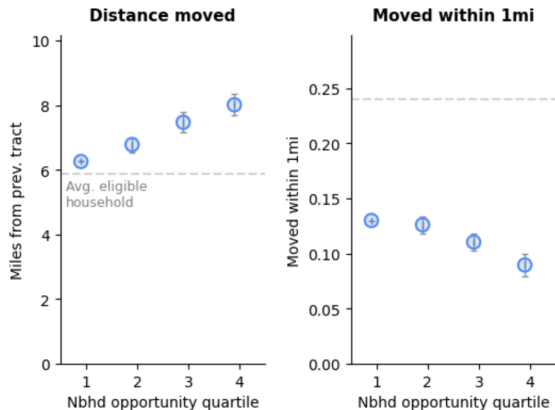
# Stepping back: what's the relevant definition of “market”?

- 1) Developers cover large areas
- Substitution may be from far away



# Stepping back: what's the relevant definition of “market”?

2) LIHTC households move from far away  
→ Rent spillovers very diffuse



Note: From Cook, Li and Binder (2026). Data cover 50 largest MSAs, 2010-2019. With MSA\*year controls.

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- Likely individually very small. Direction is theoretically ambiguous

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**Suggestion:** simple model that illustrates which parameters would affect whether the *rent* and *construction* spillovers are more/less diffuse

- Should depend on substitution patterns of both developers and households
- E.g., When is new construction more/less from outside option?

## Comment #1: ACS average rent includes LIHTC units

Observed mean rent  $R$  covers  $L$  LIHTC units renting at  $R_L$  and  $N - L$  market-rate with mean rent:

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⇒ will reduce estimated rent response if LIHTC rents are below non-LIHTC rents

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- Under-reporting in surveyed rents? Quality adjustments? Check posted rents?

## Comment #2: Defining the control group

*“Control block groups are defined as those for which these applications, and applications that occurred in any other period, were unsuccessful” (p.15)*

- Conditions on a future, perhaps endogenous outcome
  - Many rejected applicants continue to re-apply – removed here if they never succeed
  - Declining neighborhoods less likely to have future applications?
  - Better approaches to handle dynamics?
    - E.g., Cellini, Ferreira and Rothstein (2010); De Chaisemartin and d'Haultfoeuille (2026)

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- Limits power for identifying any heterogeneity

# Conclusion

## Important contribution to literature on LIHTC's equilibrium effects

- Impressive data collection. ~25k applications. Winner vs. loser approach is nice and clean
- Rare to see effects on neighborhood rents
- **Punchline:** relative to my priors, nbhd-level crowd-out is *lower* and effects on rents are *higher*

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## Opportunities to increase paper's impact

1. Teach us about what "crowd-out" means, and at what geographic scope we should care
  - How much can differences in geography/estimands reconcile findings in the literature?
2. More on heterogeneity by neighborhood, type of development, etc.
  - Is crowd-out lower in low-poverty neighborhoods? Different effects on rents?
  - Effects for senior vs general LIHTC buildings?

**Thank you!**

# Many smaller comments

- Useful to see deep dive on disagreements between Cotality/Corelogic, NHPD, and HUD
- Welfare exercise
  - Developer surplus number currently assumes no operating costs
  - Discount rate of 10% seems very high
  - Subsidy cost is currently applied to the net units, not full set of subsidized units
- Measurement issues with ACS survey rents
  - Often below LIHTC rent, even when compared to data from developers on charged rent
  - Effects on *posted* rents from other sources? (E.g., Altos, Craigslist, Zillow index)
- ACS occupancy counts may be biased proxy for total # of units if LIHTC units are more likely to be occupied than average multi-family development
  - Likely, given rationing of below-market units? Vacancy rates are very low for LIHTC

# Many smaller comments

- Geocoding rate is low ( $\sim 50\%$ )
  - If addresses have zipcodes, compare matched/unmatched balance using zipcode-level neighborhood characteristics from ACS
  - LLMs might be good at cleaning up addresses pre-geocoding
- Table 5 is not necessarily capturing turnover. Size of housing stock is changing, so there may be new entrants to neighborhood without any residents leaving
- Does winning a development affect the probability block group boundaries change?

# References

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# Rent back of envelope inputs

| Quantity                             | Source                                   | Value   |
|--------------------------------------|------------------------------------------|---------|
| $N_C$ : multifamily units, control   | Table 2A, col. 3, ctrl avg               | 223.9   |
| $\Delta N$ : new multifamily units   | Table 2A, col. 3, 2020–24                | +73     |
| $R_C$ : contract rent, control       | Table 4, col. 1, ctrl avg                | \$1,093 |
| $\Delta R$ : change in contract rent | Table 4, col. 1, 2020–24                 | –\$77   |
| $L_C$ : LIHTC units PIS, control     | Table 2A, col. 2, ctrl avg               | 33.3    |
| $\Delta L$ : new LIHTC units PIS     | Table 2A, col. 2, 2020–24                | +85     |
| $R_L^{\max}$ : contract rent cap     | Fig. 7 note; gross cap – util. allowance | \$1,125 |

$$R_T^O = \frac{(N_C + \Delta N)(R_C + \Delta R) - (L_C + \Delta L)(R_L)}{(N_C + \Delta N) - (L_C + \Delta L)} = \frac{(223.9 + 73)(1093 - 77) - (33.3 + 85)(1125)}{(223.9 + 73) - (33.3 + 85)} = \$943$$

$$R_C^O = \frac{N_C \cdot R_C - L_C \cdot R_L}{N_C - L_C} = \frac{223.9(1093) - 33.3(1125)}{223.9 - 33.3} = \$1,088$$

$$\Delta^O = 943 - 1088 = -\$145 \text{ } (-13.3\%)$$